

EVGrid AI: Smart EV Charging Network Planning System

AI & GIS-Powered Decision Support System for Optimal Infrastructure Placement & Grid Integration in Kopargaon Taluka

PROBLEM STATEMENT ID skh042	PS CATEGORY Software	THEME Transportation
PROBLEM STATEMENT TITLE EV Charging Network Planning	TEAM ID LYFWGP	TEAM NAME Team Nirdhar

 PROJECT TEAM & ROLES			
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Problem Statement & Current Challenges

ANALYSIS

PROBLEM STATEMENT

The rapid adoption of electric vehicles requires strategically planned charging infrastructure.

However, identifying suitable locations based on demand, accessibility, and grid availability remains highly challenging. Our objective is to develop a planning system that assist authorities in designing an efficient EV changing network.



CURRENT CHALLENGES IN KOPARGAON

 Limited EV charging stations around Kopargaon Taluka.	 Locations selected without AI or GIS analysis.	 No prediction of future EV charging demand.	 Limited analysis of electricity grid capacity.
 Poor coordination in infrastructure planning.	 High installation costs due to inefficient planning.	 Increasing tourist traffic to Shirdi requires better facilities.	 Dependence on fossil fuels increases regional pollution.

Proposed Solution & Key Innovation

IDEA OVERVIEW

EVGrid AI Platform Architecture

An intelligent, multi-layered decision support system analyzing 7 spatial parameters to identify optimal EV charging station locations across Kopargaon Taluka.

- ✓ **AI Location Scoring:** Automated ranking based on traffic, grid feasibility, and tourism.
- ✓ **GIS Heatmap Analytics:** Visual spatial analysis mapping high-demand pilgrimage corridors to Shirdi.
- ✓ **Demand Forecasting:** Predictive modeling for 2027–2035 EV growth patterns.
- ✓ **Budget Optimization:** Maximizes charger coverage within municipal financial constraints.

Why EVGrid AI is Unique

Replacing outdated, manual visual surveys with scientific, data-driven planning:

Integrated Solar Dashboard: Recommends hybrid solar + storage for stations with weak grid connections.

Government Decision Dashboard: Automated PDF report generator with full cost estimations & environmental impact metrics.

Scalable Framework: Designed for Kopargaon, replicable across Ahmednagar.



Existing vs. EVGrid AI Paradigm

Traditional Approach

Manual surveys, uncoordinated setup, high capital waste

EVGrid AI Approach

AI suitability scoring, GIS maps, grid load matching

Technical Approach & Architecture

METHODOLOGY

System Workflow

- 01 Data Ingestion** : GIS spatial layers, traffic census, sub-station capacities, Shirdi tourist data.
- 02 AI Evaluation Engine** : Multi-criteria weighted scoring algorithm computes site ratings (0-100).
- 03 Grid & Solar Feasibility** : Load check against MSEDCL substations + rooftop solar estimates.
- 04 Dashboard & Optimization** : Interactive GIS maps, phase-wise budget optimizer & PDF report generation.

Technology Stack

Frontend & Maps

React.js, Tailwind CSS, Leaflet.js, Mapbox GL GIS AIP

Backend & AI

Python (FastAPI / Flask), Scikit-learn, GeoPandas, NumPy

Database & Spatial

PostgreSQL with PostGIS extension for spatial querying

Analytics & Export

ReportLab PDF Generator, Optimization Solvers



AI Site Evaluation Weightage Matrix

25%

Traffic Density

20%

Population & Grid

10%

Tourism & Commercial

Feasibility & Risk Mitigation

IMPLEMENTATION STRATEGY

Technical Feasibility

- ✓ Uses standard open-source GIS libraries (PostGIS, Leaflet) ensuring zero license cost.
- ✓ Modest server requirements for Kopargaon spatial dataset processing.
- ✓ Easily integrates with government Open Data portals and MSSEDCL power grids.

Operational Viability

- ✓ Empowers Kopargaon Municipal Council with automated decision-making.
- ✓ No technical expertise needed for town planners due to simple dashboard UI.
- ✓ Accelerates EV masterplan approvals from months to days.

Financial Feasibility

- ✓ Optimizes budget allocation (e.g. ₹5 Crore target cap) to prevent capital misallocation.
- ✓ Supports PPP (Public-Private Partnership) revenue-sharing models.
- ✓ Eligible for PM E-BUS / FAME-II government grants.

Key Risks & Mitigation Matrix

Challenge / Potential Risk	Impact Level	Mitigation Strategy in EVGrid AI
Electricity Grid Instability / Capacity Limits	High	Integrated Solar Recommendation module suggests hybrid solar-battery storage for weak grid areas.
Spatial Data Integration Delays	Medium	Uses open OpenStreetMap spatial layers combined with periodic offline survey data fallback.
High Initial Infrastructure Investment	Low	Budget Optimizer prioritizes locations into Phase 1, Phase 2, Phase 3 based on ROI & immediate demand.

Impact & Regional Benefits

EXPECTED OUTCOMES

1,250

Tonnes CO₂ Saved / Year

5.6 Lakh

Liters Fuel Saved / Year

22,800+

Projected EVs Supported (2035)

82/100

Target Environmental Green
Score

Stakeholder Value Proposition

For Municipal Government: Scientific investment planning, zero wasted funds, automated PDF report generation for state approvals

For Shirdi Pilgrims and Tourists: Reliable, fast-charging corridor along highway routes connecting Kopargaon and Shirdi.

For Local Residents and Commerce: Improved air quality, reduced urban noise, and economic growth near charging hubs.



Research , Data & Future Roadmap

REFERENCES & SCALE

Demand Forecasting Model (Kopargaon)

Year	Est. EV Count	Peak Hours	Required Chargers
2027	4,560 EVs	6 PM - 10 PM	120 Chargers (Phase 1)
2030	11,250 EVs	6 PM - 11 PM	300 Chargers (Phase 2)
2035	22,800 EVs	5 PM - 12 AM	600 Chargers (Phase 3)



Scalability & Future Scope

- Expansion beyond Kopargaon to entire Ahmednagar District, Maharashtra, and Smart Cities.
- IoT Real-Time Charger Occupancy & dynamic tariff integration.
- Integration with Vehicle-to-Grid (V2G) technology for peak load balancing.

Policies & Research References

- ✓ Maharashtra Electric Vehicle Policy (2021 & updates) guidelines for public charging.
- ✓ Ministry of Power (India) Charging Infrastructure Guidelines & Standards.
- ✓ Central Electricity Authority (CEA) Spatial Grid Capacity Assessment Standards.

THANK YOU

— FOR YOUR TIME AND ATTENTION —

TEAM NIRDHAR

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"Driving the Future of Sustainable
Mobility through AI & GIS
Innovation."